

Project Report

**Supply Chain Analysis and Optimization at
Garumuni Foods (PVT) LTD**



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Contents

List of Tables.....	3
List of Figures	3
Acknowledgement	4
1. Introduction	5
1.1 Background of the Study	5
1.2 Importance of Supply Chain Management in SMEs	5
1.3 Objectives of the Study.....	5
1.4 Scope of the Report	6
2. Company Profile	6
3. Methodology.....	7
3.1 Data Collection Methods	7
3.2 Activities at the Field	7
3.3 Analysis Tools and Techniques	7
4 Supply Chain Overview	8
4.1 Key Stakeholders.....	8
4.2 Supply Chain Structure	8
4.3 Flow of Materials	9
4.4 Flow of Information	9
4.5 Flow of Finances	9
5. Supply Chain Process Analysis and Visualization Tools	10
5.1 Value Stream Mapping	10
5.2 Swimlane Diagram	11
5.3 Fishbone Diagram	12
6. Supply Chain Mapping	13
6.1 Supply Chain Detailed Diagram	13
6.2. Explanation of every step in the Supply Chain	13
7. Supply Chain Performance Analysis	15
8. Proposed Solutions and Recommendations	16
8.1 Digital Inventory Recording and Traceability System (Mobile Application)	16
8.2 Backward Integration through Own Cultivation	18

8.3 Farmer Development and Partnership Program.....	18
8.4 Project Timeline	19
10. Pilot Implementation Plan.....	20
10.1 Introduction	20
10.2 Objectives of the Pilot	20
10.3 Scope of the Pilot.....	20
10.4 Implementation Approach.....	20
.....	17
10.5 Key Performance Indicators (KPIs).....	21
10.6 Expected Outcomes	21
10.7 Risks and Mitigation.....	21
10.8 Conclusion	22
11. Evidence of field work.....	22
Letter.....	22
Questionnaire	23
Field Visit Photographs	24
References.....	24

List of Tables

Table 1: Supply Chain Performance Analysis	15
Table 2:Timeline	19
Table 3: Risks and Mitigation Plan.....	22

List of Figures

Figure1: Value Stream Map.....	10
Figure2: Swimlane Diagram	11
Figure3: Fishbone Diagram	12

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Executive Summary

1. Introduction

1.1 Background of the Study

Supply chain management is of great importance in the smooth flow of materials, information, and money in an organization. This paper aims at the supply chain analysis of Garumuni (Pvt) Ltd, a small spice exporting company located in Sri Lanka. The company which started in 1955 and is today controlled by its third generation deals with processing and exporting the cinnamon and cloves to external markets including Europe, Australia and New Zealand.

Being a third-level processor in the supply chain, the company obtains raw materials through collectors who obtain them through small-scale farmers. The major value-adding processes that the organization undertakes before the final products are exported are sorting, grading, packaging and quality assurance. Although the company has been in the industry for years, it depends on manual operations and conventional systems and this poses a number of operational challenges. This paper will be seeking to analyze these real practice and assess the effectiveness of the supply chain of the company.

1.2 Importance of Supply Chain Management in SMEs

Small and medium-scale enterprises (SMEs) are particularly susceptible to the role of supply chain management because it directly influences their competitiveness, cost-efficiency, and capacity to satisfy the needs of customers. SMEs are often limited in resources and therefore there is a need to streamline procurement, production and distribution processes. A well-functioning supply chain helps such organizations to cut down on costs, enhance product quality and offer on time delivery to customers.

In Garumuni (Pvt) Ltd case which is an export-oriented company, the supply chain efficiency is even more crucial because of the rigid international quality standards, unstable market conditions and logistical issues. Lack of proper supply chain management may cause delays, high costs, and customer mistrust. Thus, supply chain operations analysis and enhancement is essential to the long-term sustainability and development of the SMEs.

1.3 Objectives of the Study

The primary goal of this research is to examine the supply chain management of Garumuni (Pvt) Ltd and the main issues which influence its effectiveness. In particular, the paper will map the entire supply chain, stakeholders, and the movement of material, information and finances. It is also aimed at assessing the processes that are in place by applying the right tools of analysis in order to detect the inefficiencies and bottlenecks.

Moreover, the research seeks to come up with effective and feasible solutions that can enhance the performance of the supply chain, such as adoption of digital technologies to enhance data management and traceability. In this way, the research aims to make practical suggestions, which can improve the efficiency of operations and aid in decision-making in the organization.

1.4 Scope of the Report

This report will concentrate on end-to-end supply chain operations of Garumuni (Pvt) Ltd, the activities at the end-to-end include the procurement of the raw materials to the distribution of the final products. Some of the relationships analyzed are supplier relations, production processes, inventory management and export logistics. The research is grounded on primary data derived by field visits, observations, and interviews with the employees of the company.

The focus is narrowed down to the internal processes and short term external relations of the firm especially with suppliers and purchasers. Although external influences (market conditions, international logistics, etc.) are taken into account, the primary focus is made on determining the areas of operation inefficiencies and offering realistic solutions within the organization. The results and suggestions should be based on the real-life solutions that can be applied under the conditions of a small-scale business.

2. Company Profile

Garumuni (Pvt) Ltd is a small-scaled family-owned export-based business in the spice business in Sri Lanka. The company was established in the Southern Province in 1955, but is nowadays led by the third generation and has a single director administrative system. The organizational structure is made up of the Chairman, Managing Director, Finance Head, Chief Supervisor and operational employees. The current staff of the company is 29 employees, and the administrative staff comprise five employees, including logistics staff.

Garumuni (Pvt) Ltd is a company that engages in processing and export of cinnamon and cloves, with main emphasis on overseas markets like Europe, Australia and New Zealand. The company is not involved in local sales. Each month, it exports about 30,000 kg of cinnamon and 30,000 kg of cloves. Its product line consists of various types of cinnamon including cut cinnamon, cinnamon powder, quilling grades (No.1 and No.2), dark cinnamon, C4, C5 and C4N5 grades. Among them, fine cinnamon is the most expensive and highest quality.

The business is a third level processor in the supply chain where farmers and collectors do primary and secondary processing respectively. The entire sorting, cleaning, packaging and export preparation are done in house. The company is certified internationally (ISO and FSSC) and meets international standards of food safety.



3. Methodology

This research study uses a methodology that is based on primary data collection and analysis of the supply chain practices of Garumuni (Pvt) Ltd. This study primarily adopted a qualitative research methodology to understand practices, problems and processes of operations.

3.1 Data Collection Methods

Data was obtained through field visits, observations and informal interviews with staff and management. Interviews with key employees like Chief Supervisor, administrators, and labourers were conducted to obtain information on procurement, production, inventory management and export. Observations were also conducted on activities like receiving, sorting, grading, packaging, and transportation.

Secondary data was sourced from company documents like Goods Received Notes (GRN), issue notes and Excel spreadsheets for inventory and production reporting. These provided insights into the material and information flow in the company.

3.2 Activities at the Field

The research was undertaken by visiting the company on several occasions. At the site, various activities along the supply chain, such as sourcing of raw material, quality checks, production and distribution operations, were observed. The observations were centered on particular aspects of the operations to gain a comprehensive picture of the supply chain.

Observations included interactions with suppliers, production processes, inventory management procedures, and documentation processes. This allowed hands-on exposure to live operations and identification of issues and bottlenecks in the system.

3.3 Analysis Tools and Techniques

A range of tools and techniques were applied to assess the supply chain performance and improvement opportunities. Mapping of the supply chain was used to gain an understanding of the overall structure and flow of activities from suppliers to customers. Value Stream Mapping (VSM) was used to identify value and non-value adding activities, and to evaluate lead times and process efficiency.

Further, diagrams such as Swimlane diagrams and Fishbone (Ishikawa) diagrams were used to clarify process responsibilities and root causes of problems. These methods helped in analyzing bottlenecks and inefficiencies to find effective solutions.

4 Supply Chain Overview

4.1 Key Stakeholders

Garumuni (Pvt) Ltd supply chain has a number of significant stakeholders who work at various levels. Small scale farmers who grow cinnamon and cloves in areas like Ambalangoda, Kamburupitiya, Matara, Kandy and Matale are the main suppliers. These farmers serve as the first processors with harvesting and preparing cinnamon sticks and cloves. The second category of stakeholders is that of collectors or secondary dealers, who pool together products of several farmers and provide bulk quantities to the company.

In the supply chain, Garumuni (Pvt) Ltd is the manufacturer and third-level processor and value addition activities, including sorting, grading, cleaning, and packaging, are carried out. The firm has several logistics partners to carry out its transportation and export needs as opposed to relying on one service provider. The last stakeholders are the international customers who are primarily those in Europe, Australia, and New Zealand who make orders on the basis of particular quality requirements and standards.



4.2 Supply Chain Structure

Garumuni (Pvt) Ltd supply chain system is multi-tiered and linking the producers of the raw materials with the international markets. It starts with the farmers then collectors who act as the intermediaries. These collectors supply the company with raw materials, which the company buys usually on a pre-determined 8 to 12 months contract.

After receiving the materials, the company undertakes in-house processing procedures then exports the completed products. Orders are normally received through e mails and production planning is done in weekly meeting where top management prioritizes the orders. Shipment by air or air freight, based on urgency, is the last phase of the supply chain. This is a structure that guarantees a smooth flow between production and global delivery even though seasonal changes and reliability of suppliers have an impact.

4.3 Flow of Materials

The material movement commences with farmers growing and harvesting cinnamon and cloves. Once processed, i.e. peeling and making cinnamon bales, they are sold to collectors. These collectors provide the raw materials to Garumuni (Pvt) Ltd with occasional single order delivery in batches given the constraints of supplies.

When the company receives the materials, it tests them, moisture tests, and physical tests to eliminate impurities like stones, hair, or any other foreign materials. The materials are then sorted, graded and packaged in the factory. The finished products are packed as per the requirements of the buyers and ready to be exported usually in 25 kg bags. Shipment of the goods to the overseas markets is the last step and delivery time is determined by the destination.

4.4 Flow of Information

The supply chain management is mainly involved in manual and semi-digital systems that govern the flow of information within the supply chain. Buyer orders are taken primarily through email and production planning decisions are taken at weekly meetings under the guidance of the top management. WhatsApp is one of the tools to aid internal communication, as the Chief Supervisor shares daily updates on production.

The stock levels, production outputs, and supplier details are also noted in Excel sheets which are being kept by the Chief Supervisor and the company director. The movement of raw materials and production processes are tracked by using documentation like Goods Received Notes (GRN) and issue notes. Even though the said methods offer traceability, the absence of a combined digital setup restricts real-time visibility and predisposes the possibility of errors and delays.

4.5 Flow of Finances

The monetary movement in the supply chain is described by direct and relatively simple dealings. The company makes most of its payments to its suppliers, such as collectors, in cash, which enables it to achieve discounts on prices and enjoy good relationships with its suppliers. This, however, also implies that the collectors at times find themselves in financial difficulties when it comes to acquiring raw materials thus causing delays in supply.

On the customer side, the payments are usually made within 45 days of goods delivery thus a stable cash flow. In others, there are advance payments by buyers and this aids in lowering the working capital requirements of the company. Raw material purchasing, packaging and labor costs are the main factors that affect the cost structure of the company. External forces like rise in shipping and insurance expenses also affect the overall financial performance of the supply chain.

5. Supply Chain Process Analysis and Visualization Tools

5.1 Value Stream Mapping

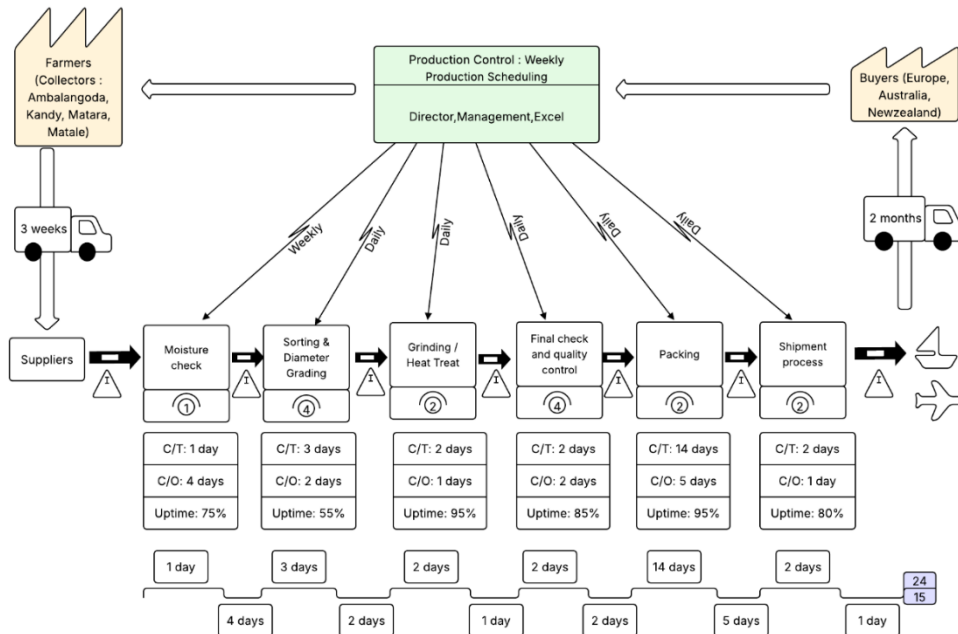


Figure1: Value Stream Map

Value Stream Map (VSM) is a lean management tool that is employed in order to map out the entire flow of information and material needed in creating and sending a product to the customer. It separates value-adding and non-value-adding activities, allowing identifying inefficiencies and waste in the process.

In the current research, VSM of Garumuni Foods (Pvt) Ltd is constructed using a typical unit of 25 kg bags of cinnamon as its packaging unit, which is the common unit that the company uses in packaging. With this unit, it is possible to measure the performance of processing, production output, and lead times in the supply chain more accurately.

The value stream starts with the procurement of the raw material through farmers and collectors to the factory and then the manual recording (GRN) and storage on the factory premises. The manufacturing process involves sorting, cleaning (destoner and hair remover machines), quality control, and filling 25 kg bags. Every step is tracked regularly, and the amount of production is frequently determined on the basis of the number of 25 kg bags produced in an hour or a day.

The processed products are then tested in laboratories and also approved by the customers and then shipped to the overseas markets like Europe, Australia and New Zealand.

The VSM points out a few areas of inefficiency such as manual inventory control, no dedicated storekeeper, inability to see inventory in real time, space constraints, and delays in supplier deliveries. All these problems lead to longer lead times, possible mistakes, and inefficiency. Moreover, there are other risks

like loss of moisture during storage and delivery of a single order that multiplies and affects the performance of the process.

The organization of the VSM into 25 kg bags allows the productivity, bottlenecks, and quantification of the inefficiencies to be easily tracked. This helps in better analysis and can be used to formulate workable solutions, including adopting a mobile-based inventory management system to enhance tracking, coordination and decision making in the supply chain.

5.2 Swimlane Diagram

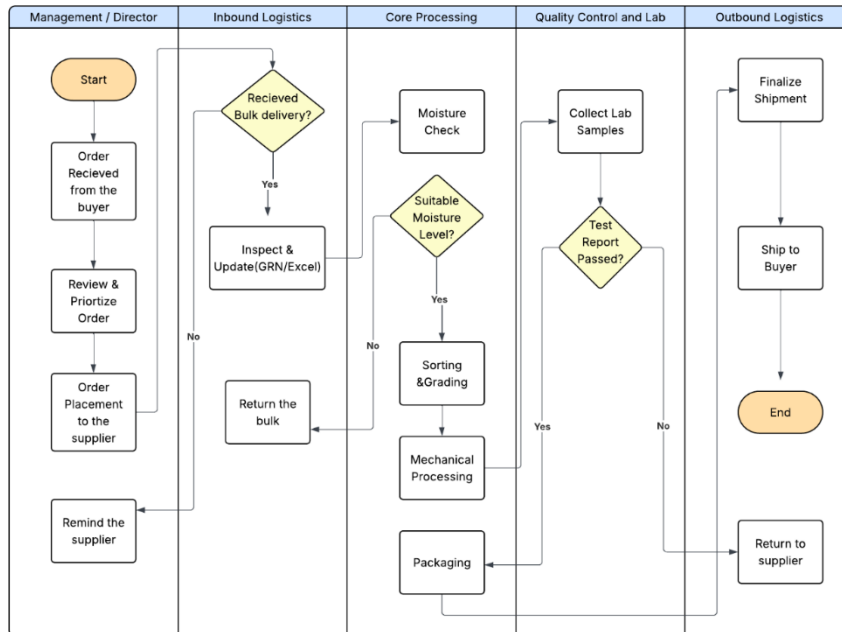


Figure2: Swimlane Diagram

Swimlane Diagram is a process mapping technique that is employed to depict the workflow by subdivision of the workload among various roles or departments. The actor is represented by each lane, and it becomes easier to comprehend who is in charge of each action in the process.

The swimlane diagram in Garumuni Foods can be used to illustrate the following major roles: Suppliers, Chief Supervisor, Employees, Management and Logistics Providers. It depicts the distribution of activities like raw material supply, receiving, quality checking, production, decision-making and shipping between these stakeholders.

As an example, the suppliers are tasked with the duty of delivering raw materials, and the chief supervisor is in charge of receiving and registering. Sorting and cleaning operations are carried out by employees, and the management has to decide and prioritize the orders. The last delivery to foreign consumers is done by the logistics providers.

The swimlane diagram assists to determine overlaps in roles, absence of clear responsibility (e.g., no storekeeper), and communication gaps. This simplifies the redesign of workflows, better allocation of responsibilities, and better coordination at the supply chain level.

5.3 Fishbone Diagram

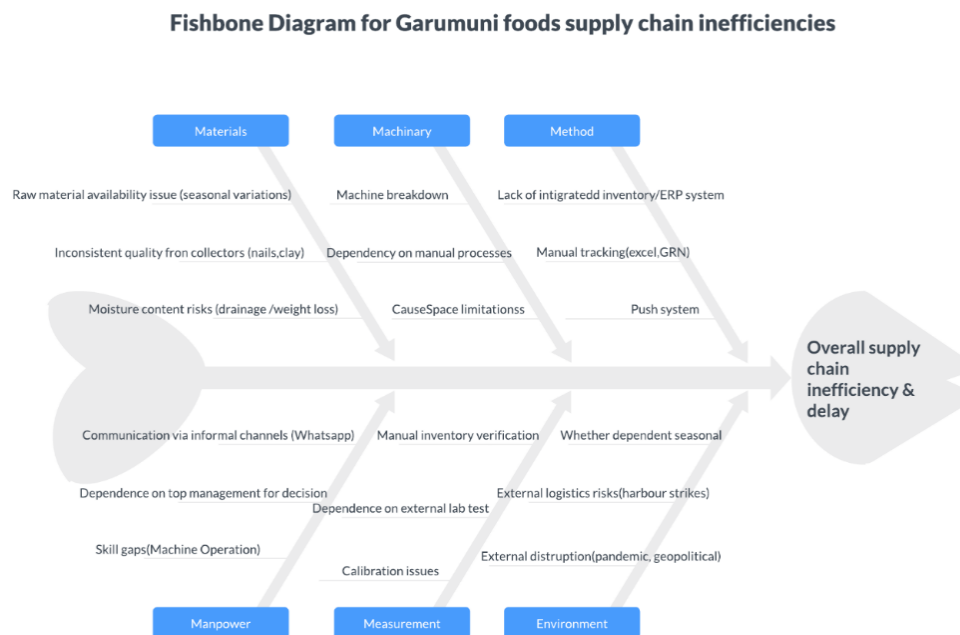


Figure3: Fishbone Diagram

The Fishbone Diagram or the Ishikawa Diagram is a method of finding and examining the root causes of a particular issue. It is a visual way of showing the contribution of various factors to a single issue to enable organizations know the root cause of a problem and not the symptoms.

The fishbone diagram may be applied to identify the issues like ineffective inventory management because of the lack of specific storekeeper and manual records in the context of Garumuni Foods (Pvt) Ltd. The causes are divided into major categories which are People, Process, Technology, Materials, Environment and Management.

As an example, in the area of people, the absence of responsibility in terms of inventory handling is possible. Process Under Process, manual Goods Received Notes (GRN) and absence of standardized procedures are the major issues. Under Technology, there is a lack of inventory management system or mobile application, which adds to inefficiencies. Others like space restriction and seasonal changes are environmental factors that impact operations.

Altogether, fishbone diagram assists in identifying root causes in a systematic manner, which contributes to the creation of effective and focused solutions.

6. Supply Chain Mapping

6.1 Supply Chain Detailed Diagram

Supply chain of Garumuni (Pvt) Ltd can be depicted as a multi-stage process which involves linking the sources of raw materials with the international customers. The flow starts with cinnamon and clove farmers who are then followed by collectors (secondary dealers), then to the company where they are processed and value added, and then to the international buyers via the logistics providers.

The general layout of the supply chain could be summed up as follows:

Farmers --> Collector - Garumuni (Pvt) Ltd (Processing & Packaging) - Logistics Providers - International Customers.

Farmers are the primary producers in this structure, and the intermediaries are the collectors, and the third level processor is the company that will enhance the quality and prepare the exports. Logistics service providers enable the supply of finished products to the international markets.

6.2. Explanation of every step in the Supply Chain

Stage 1: Raw Material Production (Farmers)

The supply chain starts with the small scale farmers who grow cinnamon and cloves. Cinnamon production involves growing, harvesting and peeling which is a specialized process that is done in Sri Lanka. Raw cinnamon is then turned into cinnamon sticks and bales by the farmers and sold. Cinnamon is a crop that requires about 3-3.5 years to produce and the production is very subjective to the season and environmental factors like rain, sunlight, and diseases of the plants.

Stage 2: Collection and Aggregation (Collectors)

Farmers sell their products to collectors or secondary dealers after first processing. These collectors collect raw materials of various farmers and distribute them in large quantities to Garumuni (Pvt) Ltd. Nevertheless, collectors can make orders repeatedly because of the shortages of supply, and their supply capability is dependent on their financial means and accessibility in the market. This step is important in connecting the small-scale producers to the company.

Stage 3: Procurement and Receiving (Company Level)

Collector purchases raw materials on purchase orders and long-term supplier contracts and sells them to the company. The Chief Supervisor and staff carry out inspections and document the details on the Goods Received Notes (GRN) upon receipt of the materials. The quality checks are performed as moisture measurement and physical inspection to detect impurities or defects. Traceability is also taken care of by the company; information about the suppliers per batch are recorded.

Stage 4: Processing and Value Addition

Raw materials are accepted and after that they go through a number of value-adding processes in the factory. These involve sorting, grading, cleaning and eliminating contaminants like stones, hair and dust.

The company employs simple machines like destoners and hair removal machines, and majority of the process is manual because of space constraints. Employees have quotas to meet on a daily basis and production is checked in every hour to ascertain efficiency and quality.



Stage 5: Packaging and Storage

Once they are processed, the final products are packed depending on the buyer specifications usually in 25 kg bags. The company maintains good hygiene and it does not lay goods on the floor but uses pallets to store goods. The lack of a specific warehouse, however, means that raw materials, as well as finished goods, are kept on-site in the factory, which poses a problem of space.



Stage 6: Quality Assurance and Testing

Quality assurance procedures such as laboratory testing of the products are done before they are exported. Some samples are also submitted to overseas laboratories e.g. in Germany to make sure that they meet buyer specifications. The products are only cleared to be shipped after approval by the buyers.

Stage 7: Distribution and Export Logistics

The last phase is the delivery of the completed products to the foreign market. Shipping is the main method of transportation that the company employs, whereas urgent orders are transported by air freight. The delivery times also vary, depending on the destination (e.g., 3540 days to Europe). The company collaborates with various logistics companies and deals with documents including bills of lading. Nevertheless, congestion at the ports, strikes, or disruptions in the world can impact the delivery schedule.

7. Supply Chain Performance Analysis

Performance Metric	Identified Issues	Impact on Supply Chain
Cost	High dependence on raw material purchasing; increased container and insurance costs; labor-intensive manual processes	Increased overall operational costs, reduced profit margins, and higher financial pressure on the company
Time (Lead Time)	Delays in supplier deliveries (1–3 weeks); partial deliveries in multiple batches; long shipping durations (e.g., 35–40 days to Europe)	Extended lead times, delayed order fulfillment, and potential customer dissatisfaction
Efficiency	Manual data recording; lack of automation; space limitations restricting machinery; repetitive processes	Reduced productivity, increased labor dependency, slower processing speed, and higher risk of human errors
Service Level	Occasional product rejections; delays due to supply and logistics issues; strict buyer requirements	Reduced customer satisfaction, risk of losing buyers, and potential additional costs (returns, rework)
Inventory Management	No formal inventory system; no real-time stock visibility; no regular physical counts; no dedicated storekeeper	Poor stock control, difficulty in planning production, risk of inefficiencies and mismanagement
Information Flow	Reliance on manual records and Excel; lack of real-time data access; delayed communication	Slow decision-making, reduced transparency, and increased chances of data inaccuracies
Quality Control	Contamination in raw materials (stones, hair, etc.); moisture variation; manual inspection processes	Increased processing time, higher labor costs, and risk of shipment rejection
Supplier Reliability	Dependence on collectors; seasonal and weather-related supply issues; financial constraints of suppliers	Supply disruptions, inconsistent material availability, and unstable production schedules

Table 1: Supply Chain Performance Analysis

8. Proposed Solutions and Recommendations

According to the supply chain analysis of Garumuni Foods (Pvt) Ltd, a number of major inefficiencies and risks were pointed out. In order to overcome these problems, three key solutions are offered, including digital transformation, stability in supplies, and improvement of supplier relationships.

8.1 Digital Inventory Recording and Traceability System (Mobile Application)

Manual data recording in the process of receiving raw materials is one of the greatest inefficiencies that have been witnessed in the company. At present, three people, including the chief supervisor, an office staff, and a security officer, individually document information like weight, supplier details, type of cinnamon and the time of receiving. After that these data will be compared by hand for error and later entered into an Excel sheet.

The process is time consuming and could be subject to human errors and led to data loss and less efficient operation. In order to address this challenge, a digital logistics and traceability system which operates on mobile is suggested. This system eliminates manual entry in favor of real time data entry using a mobile interface.

The application will enable workers to enter batch information by two workers directly into the system which will be integrated with a cloud database. Each entry is assigned with a unique batch ID that allows easy tracking of raw materials. To guarantee high accuracy, we implemented a programmatic verification layer. The system automatically executes a logical comparison between the two workers' inputs for every batch. At all the parameters are equal, the system will confirm the accuracy; otherwise, an error will be observed instantly.

The system is designed for annual longevity, meaning the company can use a single Google Sheet file for an entire year. The backend logic is programmed to detect the current date and automatically generate a new worksheet (tab) within the file for every new day of operation. This prevents any single sheet from becoming overcrowded and unreadable.

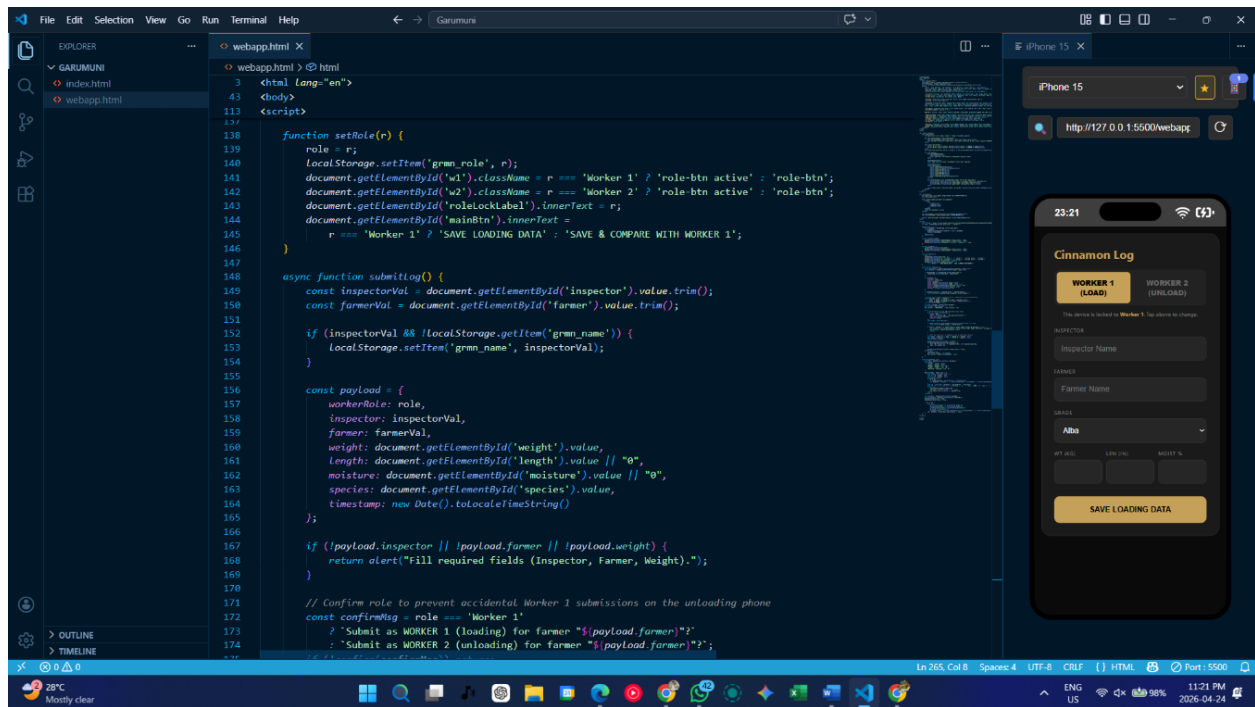
This solution can contribute greatly to accuracy of data, real time visibility and traceability and the removal of redundant manual work. It also offers a powerful stage towards future digital integration e.g. ERP systems.

[Link 1 - Access the WebView application](#)

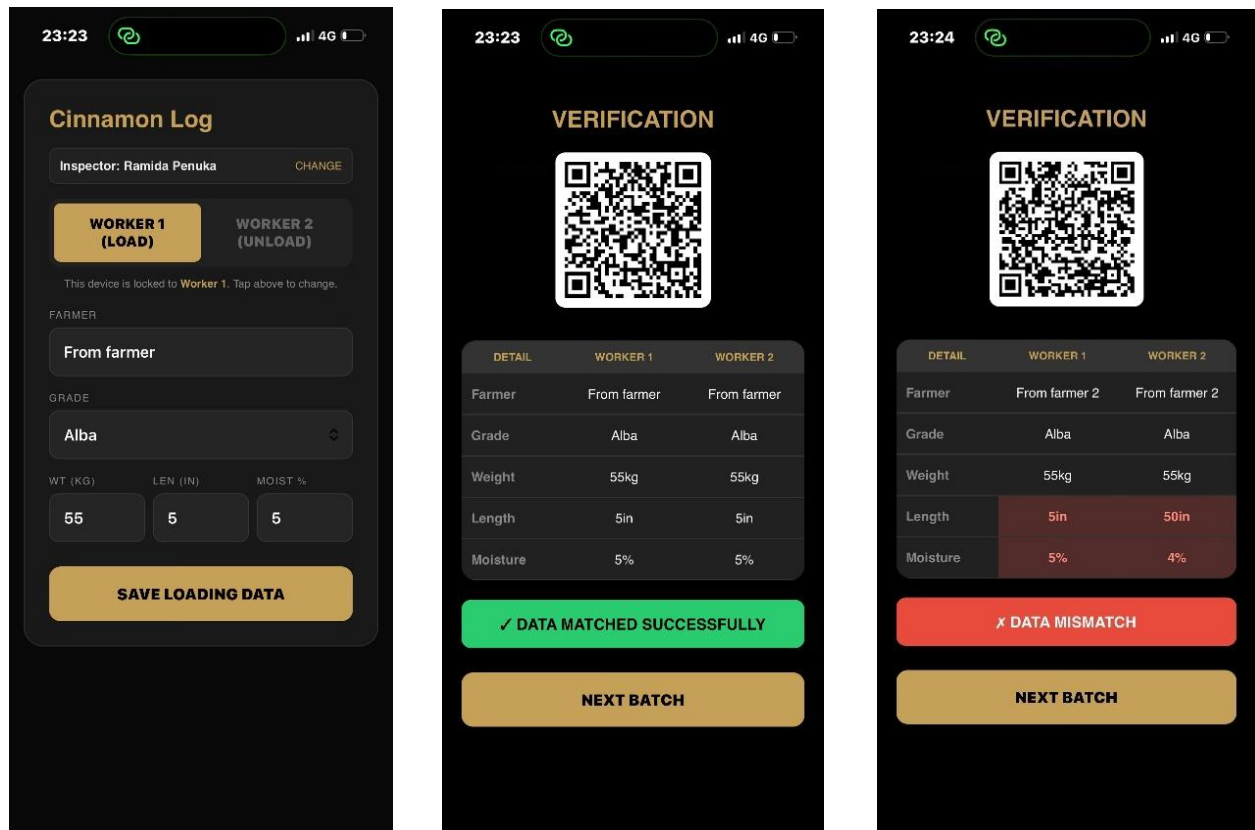
[Link 2 – Access the full code developed by the team](#)

[Link 3 - Navigate to the Pilot Project section for a full breakdown of the system.](#)

Images of VS Code development environment



Application Screen Shots



Screenshot of the Excel Data Base

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	B-12-57-10	Ramida Penuka	Ramida Penuka	12:57:05	Alba	50	5	5	5						PENDING	
2	B-12-59-22	Gg	Ramida Penuka	12:59:19	Alba	50	5	5	5	Oshan Silva	12:59:43	Alba	50	5	5	MATCHED
3	B-13-00-35	Gg	Ramida Penuka	13:00:31	Alba	50	5	5	5	Oshan Silva	13:00:32	Alba	50	5	10	MISMATCH
4	B-13-01-23	Gg	Ramida Penuka	13:01:20	Alba	40	4	4	4	Oshan Silva	13:01:26	Alba	40	4	4	MATCHED
5	B-13-01-48	Gg	Ramida Penuka	13:01:45	Alba	50	5	5	5	Oshan Silva	13:01:53	C5 Special	50	5	5	MISMATCH
6	B-13-02-28	Gg	Ramida Penuka	13:02:24	M4	50	5	5	5	Oshan Silva	13:02:27	M4	50	5	5	MATCHED
7	B-13-22-05	Gm	Oshan Silva	13:22:00	Alba	50	5	5	10	Ramida Penuka	13:22:52	Alba	50	5	10	MATCHED
8	B-13-23-32	Gg	Oshan Silva	13:23:28	C5 Special	50	5	5	10	Ramida Penuka	13:24:10	M5	50	5	10	MISMATCH
9	B-13-25-16	Gg	Oshan Silva	13:25:11	Alba	50	5	5	5	Ramida Penuka	13:25:17	Alba	50	5	5	MATCHED
10	B-13-37-01	Farmer	Oshan Silva	13:36:58	Alba	50	5	5	5	Ramida Penuka	13:37:59	Alba	50	8	5	MISMATCH

8.2 Backward Integration through Own Cultivation

The other significant weakness that the company is experiencing is excessive reliance on external suppliers such as farmers and collectors. This dependence frequently results in delays in the delivery of raw materials, fluctuation of prices and quality, particularly during off-seasons or bad weather.

In order to reduce this risk, it is advisable that Garumuni Foods grows gradually towards backward integration by planting their own cinnamon farms.

Creating their supply base will enable the company to have a stable and reliable source of raw materials, have the same level of quality and the company will have less dependability on the intermediaries. Though this costs, in the short term, in investment and time (cinnamon would grow around 3-3.5 years), it is beneficial in the long term because of the ability to control costs, secure supply, and enhance bargaining power.

Such a strategy also allows higher planning and minimizes the uncertainty in production scheduling.

8.3 Farmer Development and Partnership Program

Along with the inner cultivation, it is crucial to enhance the relations with current suppliers. Today, the company is experiencing problems with delayed deliveries, financial inability of collectors, and unreliable supply.

To overcome this, it is suggested to introduce Farmer Development and Partnership Program. This would be a program that aims at establishing long term relationships with the farmers on training, support and collaboration.

The company will be able to conduct training on the best agricultural practices, quality standards, moisture control as well as sustainable farming practices. Also, technical advice and potential financial aid (advance payments or resources aid) can enhance the ability of farmers to produce a steady and quality raw material.

This kind of a program improves supplier reliability, consistency in quality and trust besides helping in sustainability of the supply chain. It reinforces the company position by establishing a more stable supplier network that is cooperative.

8.4 Project Timeline

Task	week 1	week 2	week 3	week 4	week 5	week 6	week 7	week 8	week 9	week 10
Identify and select suitable SME; obtain permission										
Plan project approach and schedule initial visit										
Conduct first visit and understand business operations										
Identify supply chain structure and collect preliminary data										
Analyze supply chain processes (procurement to distribution)										
Develop supply chain map										
Conduct visit two and observe operations and identify inefficiencies										
Validate problems with management										
Develop Fishbone diagram and identify root causes										
Propose and evaluate feasible solutions										
Design mobile application workflow and data structure										
Develop backend and create prototype										
Conduct third visit and present solution and demonstrate system										
Conduct test run and collect feedback										
Fix issues and improve functionality/UI										
Finalize system and prepare documentation										
Conduct fourth visit and deploy system and provide user training										
Hand over completed solution										
Compile analysis and prepare diagrams										
Complete and submit final report										

Table 2:Timeline

10. Pilot Implementation Plan

10.1 Introduction

A Digital Inventory Recording and Traceability System, including a mobile app, is suggested to overcome problems with the current manual inventory recording system. The purpose of the pilot is to implement and test the system in a real-world setting, assess its performance, and make improvements prior to full deployment.

10.2 Objectives of the Pilot

The pilot trial aims to:

- To eliminate any errors in data recording
- To enhance data accuracy and integrity
- To reduce time for data recording and verification
- To improve traceability of raw materials
- To assess the users' acceptability of the system

10.3 Scope of the Pilot

The pilot will concentrate on the receiving of the raw material, in particular the 25 kg bags of cinnamon. It will involve: Chief Supervisor Office Staff Member Security Officer. The pilot will replace the existing manual recording and verification by the three individuals.

10.4 Implementation Approach

The pilot will be implemented in stages, with minimal impact on the existing operations.

Phase 1: Preparation Study existing manual process Determine the data fields (supplier, weight, date, type of cinnamon) Select devices and users

Phase 2: System Setup Install mobile application Set up database (Google Sheets/backend) Set user roles and permissions

Phase 3: Training Train users on system Show how to enter and validate data Explain system-generated outputs

Phase 4: Pilot Execution Operate system in real-time At first parallel to manual system Progressively record only electronically

Phase 5: Monitoring and Evaluation Track system performance Collect user feedback Identify issues and improvements

10.5 Key Performance Indicators (KPIs)

The pilot will be assessed based on:

1. Reduction in data entry time
2. Reduction in recording errors
3. Accuracy of data matching
4. User satisfaction
5. System reliability

10.6 Expected Outcomes

- Reduced double manual entry
- Improved efficiency and accuracy
- Better traceability and visibility
- Real-time data for improved decision making Increased operational efficiency

10.7 Risks and Mitigation

Risk	Likelihood	Impact	Severity	Mitigation Action
User Resistance to New System	High	Medium	High	Provide proper training, awareness sessions, and hands-on demonstrations to employees before implementation
Lack of Technical Skills Among Workers	High	Medium	High	Design a simple and user-friendly interface and offer continuous support during the pilot phase
Data Entry Errors During Initial Stage	Medium	Medium	Medium	Implement validation checks, duplicate entry verification, and cross-checking mechanisms in the app
System Downtime / Technical Failures	Low	High	Medium	Use reliable cloud services, maintain backups, and ensure basic IT support availability
Poor Internet Connectivity	Medium	Medium	Medium	Enable offline data entry with later synchronization when connectivity is available
Data Security Risks	Low	High	Medium	Use secure login systems, role-based access, and encrypted cloud storage
Integration Issues with Existing Processes	Medium	Medium	Medium	Run the digital system parallel with the manual system during the pilot phase for smooth transition
Increased Initial Cost	Medium	Medium	Medium	Start with a small-scale pilot project and scale gradually based on results

Incorrect or Incomplete Data Migration	Low	Medium	Low	Verify and clean existing data before entering into the system
Lack of Management Support	Low	High	Medium	Ensure active involvement of top management and regular monitoring of pilot progress

Table 3: Risks and Mitigation Plan

10.8 Conclusion

The Digital Inventory Recording and Traceability System pilot is expected to enhance the raw material receiving process's efficiency and accuracy. The pilot experience will be used to improve the system and roll it out to the entire operation.

11. Evidence of field work

Letter



Garumuni Foods (Pvt) Ltd

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To Whom It May Concern,

This is to certify that the students from the University of Moratuwa have conducted a supply chain management project in collaboration with Garumuni Foods (PVT) LTD, a company engaged in Sri Lankan Spice export operations.

As part of the project, the students visited our organization on four occasions. During these visits, they closely observed our operational processes and gathered relevant supply chain data to support their academic analysis.

On their final visit, the students presented a set of proposed solutions addressing current supply chain challenges within our organization, particularly in areas related to sourcing, processing, and logistics.

The project was carried out with the guidance and support of Mr. Thusith Pinnadoowe Head of Finance of Garumuni Foods (PVT) LTD. We confirm that the analysis and recommendations presented are based on the students' independent academic work conducted with our cooperation.

We appreciate their professionalism, commitment, and structured approach throughout the engagement and wish them success in their future endeavors.

Yours faithfully,

Garumuni Foods (PVT) LTD


G.A.C. De Silva
DIRECTOR

Questionnaire

Structured questionnaires served as one of the main sources for gathering specific data related to the workings of the company's supply chain system. The questions in the questionnaires focused on several areas, including procurement, production, inventory management, quality control, logistics, and other supply chain issues.

Data was collected via discussion with the company's finance head directly. The data thus gathered proved to be quite useful and practical in terms of its accuracy and relevance to the company's operations. All gathered data was then put to use and included in the appropriate chapters in the report.

Section A: Company & Operations

6. Can you provide a brief overview of the company and its core operations?
7. What are the main products produced and exported by the company?
8. What is the approximate production capacity (daily/monthly)?
9. Are there any seasonal variations in production?

Section B: Suppliers & Procurement

5. Who are your main suppliers and sourcing regions?
6. How do you select and evaluate suppliers (e.g., price, quality, reliability)?
7. What is the typical lead time for receiving raw materials?
8. Do you experience delays in supplier deliveries? If so, what are the reasons?
9. Are there long-term contracts with suppliers?

Section C: Production & Processing

10. Can you briefly describe the main processing steps from raw materials to finished products?
11. Which processes are manual and which are automated?
12. Are there any bottlenecks or delays in production?
13. How are defective or rejected materials handled?

Section D: Inventory Management

14. How is inventory managed and recorded in the company?
15. Are there any challenges related to stock management or storage?
16. How do you handle stock shortages or excess inventory?

Section E: Quality Control

17. How do you ensure product quality and safety?
18. Are lab tests or certifications required before exporting?

19. How do you handle customer complaints or product rejections?

Section F: Distribution & Logistics

20. How are finished products delivered to international customers?

21. What are the typical delivery times for major export markets?

22. Do you face challenges in shipping or logistics?

Section G: Information & Financial Flow

23. How are customer orders received and processed?

24. How is production and inventory information recorded and shared?

25. What are the payment terms with suppliers and customers?

Section H: Challenges & Improvements

26. What are the main challenges in your supply chain?

27. Which processes cause the most inefficiencies?

28. Are there any plans to improve operations or adopt new technologies?

Field Visit Photographs



